

- Question 4:
 - Let n = number of nodes in the graph
 - Time Complexity is $O(n^2)$ because BFS visits each node at most once. Each node that is removed, every neighbor is checked, which is n possible neighbors for each node. Up to n nodes visited and n possible neighbors for each node makes it $n * n = n^2$.
 - Space Complexity is $O(n)$ because visited array is $O(n)$ space and queue is $O(n)$ space. Making the total space complexity $O(n)$.
- Question 5:
 - Let n = number of nodes in the graph
 - Time Complexity is $O(n^4)$ because the program has four nested loops and each loop runs n times. Making it $n * n * n * n = n^4$. The keyKey method only processed 4 nodes, making it constant time, thus the final complexity is $O(n^4)$.
 - Space Complexity is $O(n^4)$ because printed cycle keys must be stored within the cycles. Bringing the final complexity to $O(n^4)$